

FDE Data Foundations Assignment

Classes 4–8 | From Client Data to a Dependable Pipeline

Assignment objective: Take a realistic operational problem from messy source data to a small, explainable, dependable data pipeline that produces trustworthy business metrics.

Problem statement

You are the FDE assigned to a client whose operations span multiple systems. Leadership has a business problem, but the data is fragmented, inconsistent, and not yet organised around the workflow. Your task is to identify the right sources, retrieve the required data, validate it, model the business workflow, define useful metrics, and build a simple repeatable pipeline that produces those metrics reliably.

Choose one project track

Track	Example problem
A — FlashEats-style	Late deliveries are hurting customer experience. Reconstruct the order workflow, identify where delay accumulates, evaluate operational interventions, and produce a dependable KPI pipeline.
B — NYC TLC	Use NYC taxi trip data to build a workflow-oriented view of trips, validate trip-duration/location data, define 3–5 operational metrics, and automate a repeatable monthly metric pipeline.
C — Your own	Choose a real-life operational problem with multiple data sources, e.g. clinic appointments, e-commerce fulfilment, campus transport, service tickets, logistics, subscriptions, or another domain you can explain clearly.

What your submission must demonstrate

Class	Skill	Evidence in your project
4	Understand sources	Map business questions → required information → source systems. Identify source ownership, grain, and any important gaps.
5	Retrieve data	Use at least two retrieval modes across SQL, API, JSON/CSV/files. Show how you know retrieval is complete and preserve raw inputs.
6	Profile & validate	Profile the data, identify meaningful quality issues, define business-oriented validation rules, and record assumptions/limitations instead of silently fixing them.

7	Model workflow	Represent entities, events/states, interactions/interventions/outcomes. Build a simple relational or event model and calculate 3–5 metrics linked to the project KPI.
8	Dependable pipeline	Turn the analysis into a repeatable pipeline: ingest → validate → transform/model → metric output. Include logging/checks, rerun behaviour, and failure handling.

Submission package

- GitHub project URL (required): submit a public or otherwise accessible repository containing the complete project.
- README.md in the repository: problem, users/stakeholders, project KPI, source overview, setup/run instructions, and what decision the output supports.
- Source map + workflow/data model diagram, stored in the repository (simple is fine).
- Code/notebook(s) showing retrieval, validation, modelling, joins/aggregations, metrics, and a runnable pipeline/script that reproduces the final output from raw inputs.
- Final evidence table/dashboard with 3–5 metrics, plus a brief Known / Unknown / Assumption / Limitation section.
- A 3–5 minute demo using the GitHub project, explaining one important FDE judgement call you made.

What matters most

We are not grading project size or UI polish. We are looking for evidence that you can move from messy client data to a trustworthy operational model and a repeatable output. Your choices should be explicit, defensible, and connected to the business KPI.

Source reasoning	Retrieval	Validation	Workflow + metrics	Pipeline dependability
20%	20%	20%	20%	20%

Think like an FDE: the goal is not “I analysed a dataset.” The goal is “I built a trustworthy path from client systems to a business decision.”